# C Kaustubh

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## Education

**B.Tech in Computer Science & Engineering** at PES University 2023 – 2027
CGPA: 8.59 **Dr. CNR Rao Scholarship** (Sem 1) **DAC Distinction Scholarship** (Sem 2 – Sem 5)

## Experience

**AI/ML Intern** @ **AVEVA (Schneider Electric)** Jun '26 – Present
• Took **Calculation Copilot** from concept to a live demo: built a fully agentic assistant within **AVEVA Connect** that turns natural language into validated KPI calculations (OEE, Availability, MTTR, MTTF) via **Azure OpenAI**, with adaptive one-shot/guided flows and multi-step tool use, replacing manual form-based authoring.
• Deployed and operated it on **Kubernetes**, independently resolving production issues, and presented it in an internal engineering knowledge-sharing session.

**Research Intern** @ **Center for Information Security, Forensics and Cyber Resilience (C-ISFCR)** Jun '25 – Jul '25
*Under Prof. Preet Kanwal*
• Designed multi-modal pipelines (YOLOv11 + Optical Flow + LLaVA-Next + LLaMA-3) for unsupervised video anomaly detection on the NWPU Campus Dataset (16.5h footage), reaching **75.48%** accuracy (F1 **0.845–0.895**); built a hierarchical LLM summarization engine benchmarked against Gemini-2.5-Flash at lower serving cost.

## Leadership & Activities

**Tech Lead @ Shunya PESU** — Built and maintain the club website in React/TypeScript/Tailwind. Jan '26 – Present

## Open Source Contributions

- **LanceDB** — Fixed Node.js process hang caused by unreleased **Napi::ThreadSafeFunction** in the RRF reranker; merged into `python-v0.32.0`.
- **Firecracker** — Fixed mutex deadlocks triggered by **clippy-tracing** instrumentation in the VMM; merged into `main`.

## Skills & Technologies

**Languages & Frameworks:** Python, Rust, C/C++, Go, TypeScript, Next.js, PyTorch, Bash
**Infrastructure & DevOps:** Docker, Kubernetes, GitHub Actions (CI/CD), Linux, Git, Redis, Kafka, Spark
**ML & Data:** PyTorch, CUDA, OpenCV, Sentence-BERT, YOLOv11, LLaVA, LLaMA, LoRA, GRPO

## Projects

- **Towards Fair Indian English ASR** *(Research, under Dr. B.Das)* — Fairness pipeline (ΔDP, ΔEO) for Whisper, Wav2Vec2, HuBERT across 19 Indian accents; **GRPO** RL post-training on Whisper + LoRA reduced **WER** to **16.5%** and shifted DP gap from systematic to non-systematic.
- **UncertaRAG** — Uncertainty-aware RAG evaluation framework benchmarking Mistral-7B across 12 quantization conditions (FP16 vs INT4). Key result: a no-retrieval SemanticVar baseline outperforms all retrieval-augmented variants on AUROC (0.815 under INT4).
- **Rakshak-AI** — An agentic compliance readiness copilot that helps startups and regulated companies become audit ready. 1st Runner-up at **GDG Cloud Bengaluru Deployfest 2026**
- **Qwen_Cuda** — Raw CUDA inference for the Qwen Model family. Full hardware acceleration with custom kernels. Beats transformers on Qwen3:8B Q4_K by +3 tok/s.
- **LocalLens** — MCP-native multimodal file search. Indexes files with **jina-embeddings-v5-omni-small**, stores in **LanceDB**, and exposes results via REST API and MCP server for direct agent and IDE integration.
- **redis-from-scratch** — Redis-compatible in-memory key-value store in Go with full RESP3 protocol support and Pub-Sub. Deployed as backing store for a REST API-based Kafka clone. CI via GitHub Actions.
- **sql-engine** — SQL database engine built in C++17: RDP Parsing, Volcano iterator execution, BTree indexing , three join algorithms (nested-loop, hash, index NLJ) with rule-based optimizer and predicate pushdown, disk persistence.